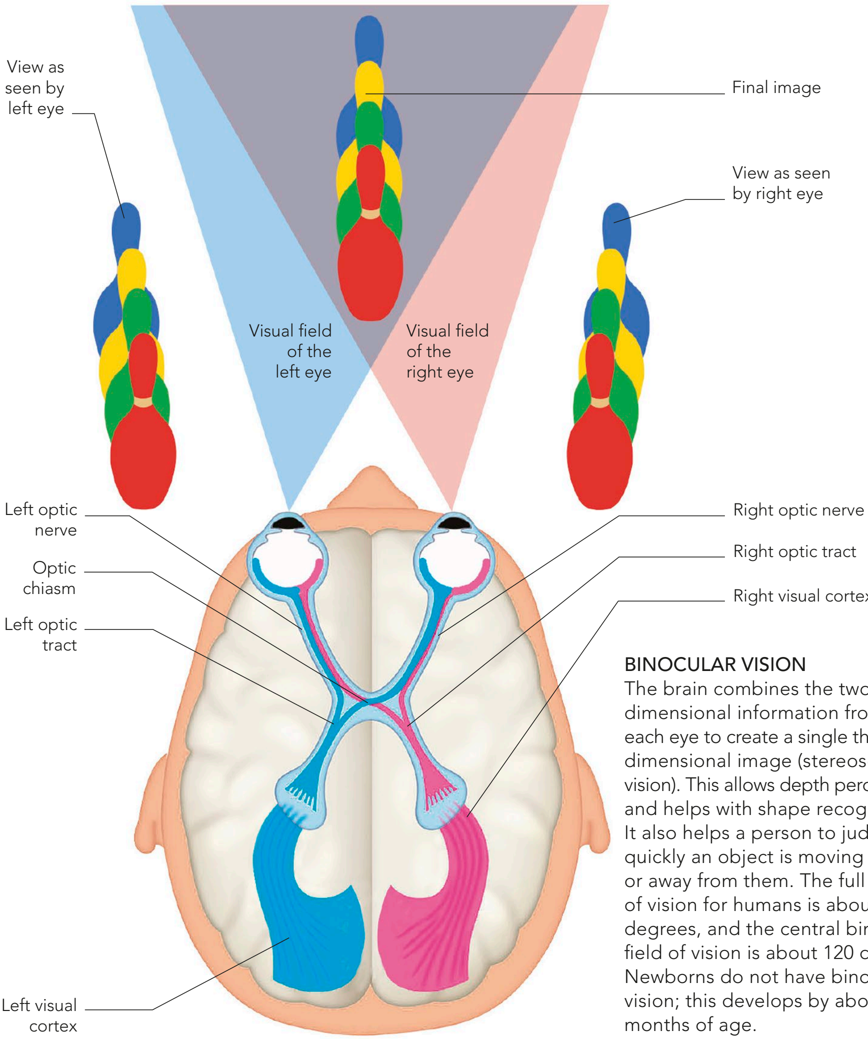
VISUAL PATHWAYS

Nerve signals conducted along the left and right optic nerves converge at a crossover junction, called the optic chiasm, at the base of the brain. Here, fibres carrying signals from the left side of each retina join and proceed as the left optic tract to the left visual cortex at the back of the brain. Likewise fibres from the right side of each retina form the right optic tract and go to the right visual cortex. Because the eyes are set apart, each sees a slightly different view of an object. The combination of the views of both eyes into a single image is called binocular vision.



BINOCULAR VISION

The brain combines the two-dimensional information from each eye to create a single three-dimensional image (stereoscopic vision). This allows depth perception and helps with shape recognition. It also helps a person to judge how quickly an object is moving towards or away from them. The full field of vision for humans is about 180 degrees, and the central binocular field of vision is about 120 degrees. Newborns do not have binocular vision; this develops by about four months of age.